

Data-Driven Discovery of Accretion Disk Hydrodynamics

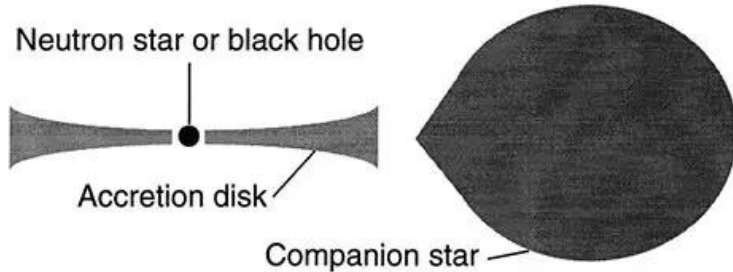
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Advanced Topics in Physics of Data

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01 Physical Context

Accretion Disk Geometry



- Matter in Keplerian orbit loses angular momentum through turbulent shear stress and flows inward.
- Outward transport of angular momentum enables mass accretion onto the central compact object.

Target Equation

$$\frac{\partial \Sigma}{\partial t} = \frac{3}{R} \frac{\partial}{\partial R} \left[R^{1/2} \frac{\partial}{\partial R} \left(\nu \Sigma R^{1/2} \right) \right]$$

Conservation laws yield:

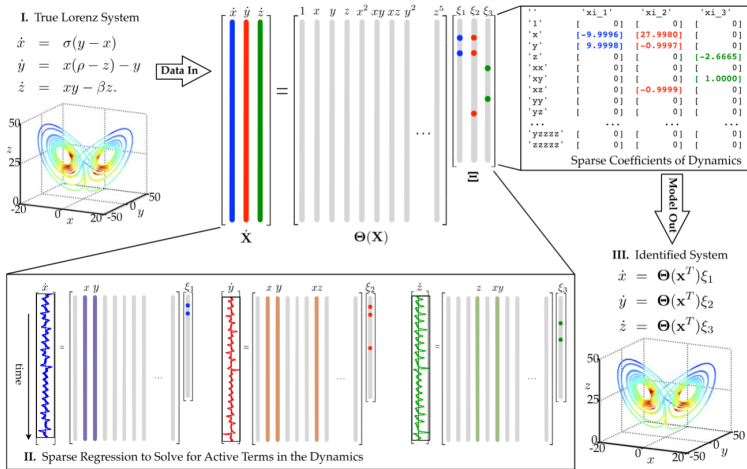
$$\partial_t \Sigma = 3\nu \Sigma_{RR} + \frac{9\nu}{2R} \Sigma_R$$

- Viscous diffusion: $3\nu \Sigma_{RR}$
- Inward geometric drift: $\frac{9\nu}{2R} \Sigma_R$

- **Black Hole:** Zero torque at ISCO
($R_{\text{in}} = 6M$) $\implies$ matter sink
 $\Sigma(R_{\text{in}}, t) = 0$.
- **Neutron Star:** Solid stellar
surface sustains torque $\implies$
matter pile-up $\Sigma(R_{\text{in}}, t) > 0$.

02 Discovery Methodology

SINDy Scheme



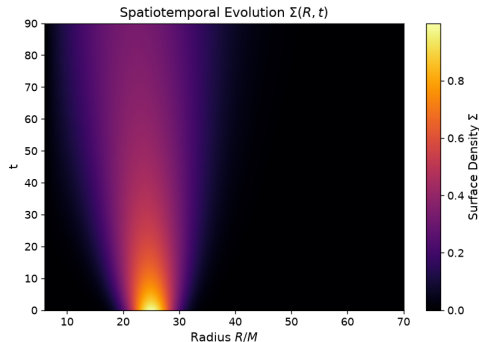
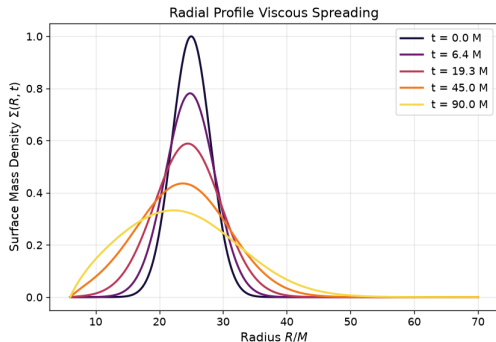
WSINDy Framework

- Differentiating noisy data amplifies noise.
- The weak formulation of PDE averages over local windows using smooth test functions $\psi(R, t)$ that vanish at boundaries.
- The advantage of it is that it moves derivatives onto the test function:

$$\int \partial_t \Sigma \psi = - \int \Sigma \partial_t \psi$$

03 Numerical Results

Forward Simulation



Hydrodynamic Evolution

Viscous Spreading

- Initial Gaussian ring centered at $R_0 = 25M$ with width $w = 3M$.
- Inward viscous drift carries matter toward the ISCO at $R = 6M$.
- Outward expansion conserves total system angular momentum.

Spatiotemporal Field

- Surface density field $\Sigma(R, t)$ discretized on $N_R = 480, N_t = 701$.
- Domain spans $R \in [6.0, 70.0] M$ and $t \in [0, 90.0] M$.
- Ground truth kinematic viscosity: $\nu = 0.1500$.

Discovery

SINDy in (R, t)

Σ_{RR} coefficient 3ν :

$$3\hat{\nu} = 0.420 \implies \hat{\nu} = 0.140$$

- Parameter error: 6.7%.
- Spurious terms (6 total).
- FVU ≈ 0.017

WSINDy in (R, t)

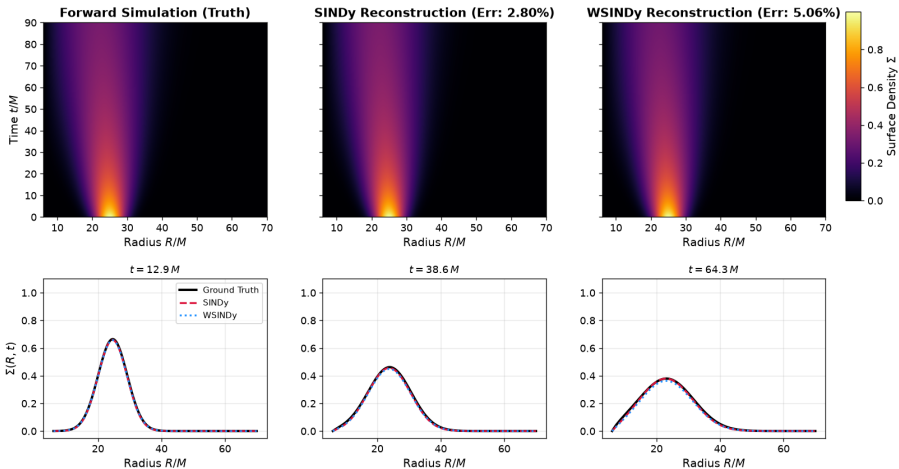
Σ_{RR} coefficient 3ν

$$3\hat{\nu} = 0.458 \implies \hat{\nu} = 0.153$$

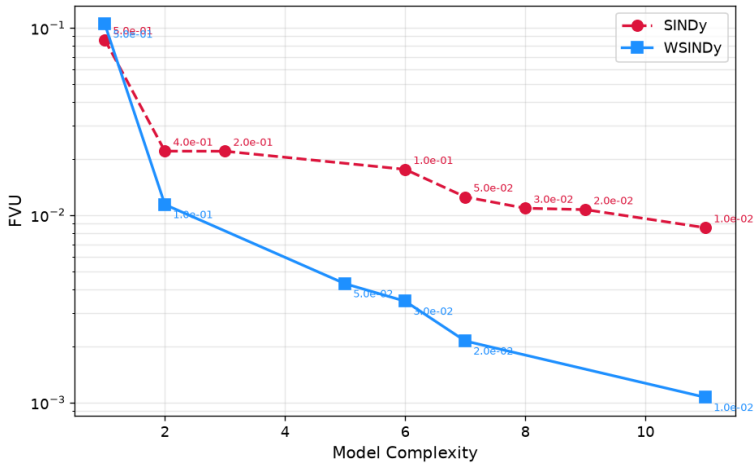
- Parameter error: 1.7%.
- 2 terms
- FVU ≈ 0.008

$$\partial_t \Sigma = 0.462 \Sigma_{RR} + 0.031 \Sigma_R$$

Out-of-the-box Reconstruction



Model Selection



Model Selection

Complexity Sweep

- Sparsity threshold swept over $\lambda \in [0.5, 10^{-4}]$.
- Fraction of Variance Unexplained:

$$\text{FVU} = 1 - R^2 = \frac{\|\mathbf{U}_t - \mathbf{\Theta}\boldsymbol{\xi}\|_2^2}{\text{VAR}(\mathbf{U}_t)}$$

- Evaluates trade-off between model complexity and fitting residual.

Pareto-knee models

Both approaches discover a 2 term equation of the form: $\partial_t \Sigma = c_1 \Sigma_R + c_2 \Sigma_{RR}$.

Sindy

$$\partial_t \Sigma = 0.448 \Sigma_{RR} + 0.0326 \Sigma_R$$

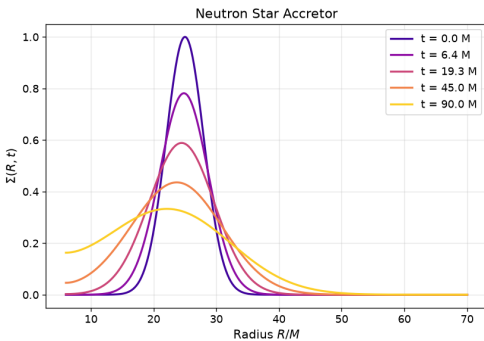
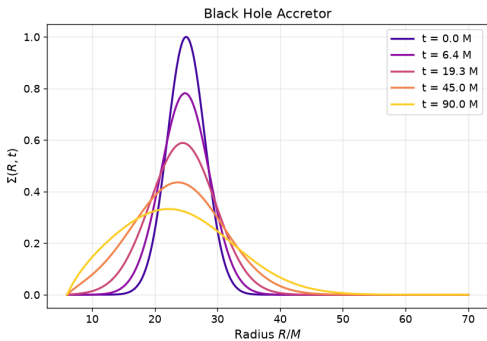
With a relative error of 0.44% and a FVU of 2.19×10^{-2} .

WSINDy

$$\partial_t \Sigma = 0.458 \Sigma_{RR} + 0.031 \Sigma_R$$

With a relative error of 1.76% and a FVU of 8.65×10^{-3} .

Compact Object Discrimination



Boundary Classification

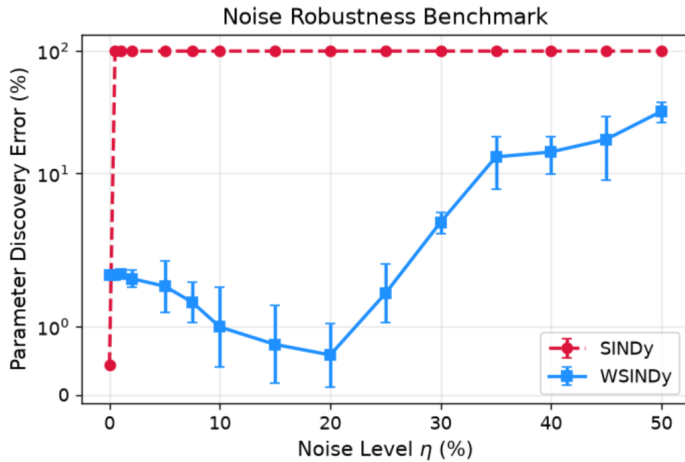
Black Hole Boundary

- Boundary condition is that ISCO orbit acts as mass sink.
- Mean inner density:
 $\langle \Sigma(R_{\text{in}}) \rangle = 0.0.$

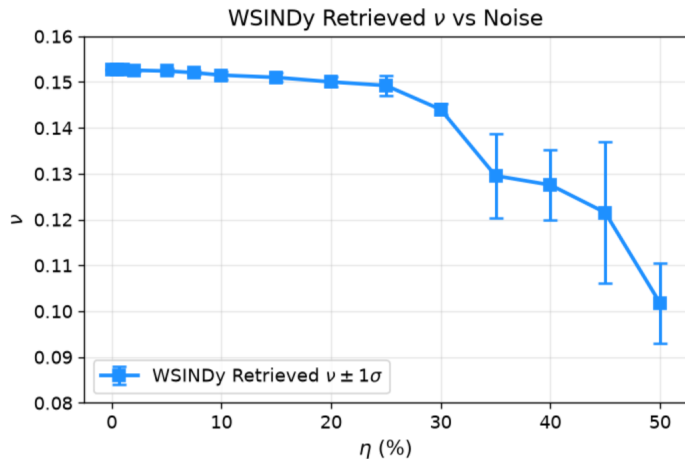
Neutron Star Boundary

- Solid surface sustains viscous torque.
- Mean inner density:
 $\langle \Sigma(R_{\text{in}}) \rangle \approx 0.06.$
- Peak inner density:
 $\max(\Sigma(R_{\text{in}})) \approx 0.16.$

Noise Robustness



Noise Robustness



Noise Benchmark

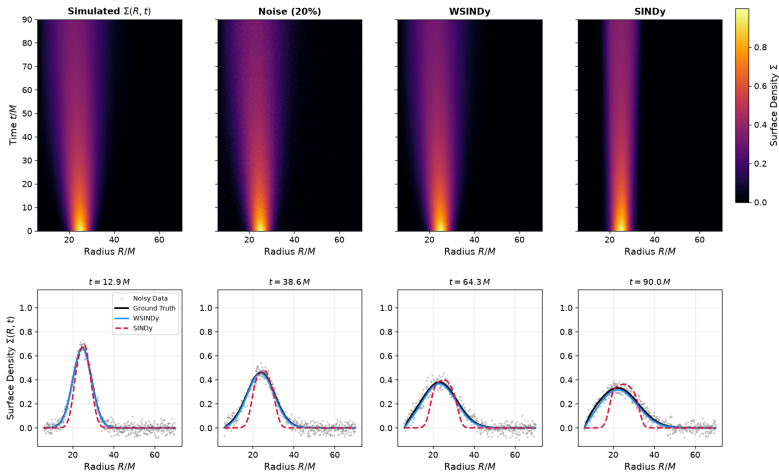
SINDy Breakdown

- Pointwise finite differences amplify high-frequency measurement noise.
- Parameter estimation error reaches 100% at noise levels $\geq 0.5\%$.
- Standard derivative evaluation fails under noisy observational data.

WSINDy Robustness

- Test-function integration by parts acts as an intrinsic low-pass filter.
- Maintains stable parameter recovery across most noise levels $\eta \in [0\%, 30\%]$.
- Error remains bounded even under 30.0% noise.

Reconstruction



04 Conclusions

Key Findings

PDE Discovery

- SINDy and WSINDy successfully identify the accretion diffusion PDE from fluid data.

Physical and Noise Stability

- Inner boundary density $\Sigma(R_{\text{in}}, t)$ separates Black Holes from Neutron Stars.
- WSINDy maintains parameter accuracy under up to 30% Gaussian noise.

Thanks!

Summary of results

Noise η	SINDy			WSINDy		
	$\hat{\nu}$	Rel. Err. (%)	k	$\hat{\nu}$	Rel. Err. (%)	k
0.0%	0.1400 ± 0.0000	$6.66\% \pm 0.00\%$	6.0	0.1526 ± 0.0000	$1.76\% \pm 0.00\%$	2.0
0.5%	0.0000 ± 0.0000	$100.00\% \pm 0.00\%$	4.8	0.1526 ± 0.0000	$1.76\% \pm 0.02\%$	2.0
1.0%	0.0000 ± 0.0000	$100.00\% \pm 0.00\%$	5.0	0.1527 ± 0.0001	$1.77\% \pm 0.04\%$	2.0
5.0%	0.0000 ± 0.0000	$100.00\% \pm 0.00\%$	5.8	0.1524 ± 0.0006	$1.60\% \pm 0.38\%$	2.0
10.0%	0.0001 ± 0.0000	$99.96\% \pm 0.02\%$	7.8	0.1515 ± 0.0009	$1.00\% \pm 0.59\%$	2.0
15.0%	0.0001 ± 0.0000	$99.96\% \pm 0.01\%$	9.2	0.1510 ± 0.0010	$0.74\% \pm 0.57\%$	2.0
20.0%	0.0000 ± 0.0000	$99.97\% \pm 0.01\%$	9.0	0.1500 ± 0.0011	$0.59\% \pm 0.47\%$	2.0
25.0%	0.0000 ± 0.0000	$99.97\% \pm 0.00\%$	9.2	0.1492 ± 0.0022	$1.49\% \pm 0.43\%$	2.0
30.0%	0.0000 ± 0.0000	$99.98\% \pm 0.00\%$	10.2	0.1440 ± 0.0012	$4.00\% \pm 0.78\%$	2.0
40.0%	0.0000 ± 0.0000	$99.98\% \pm 0.01\%$	9.4	0.1275 ± 0.0078	$14.97\% \pm 5.17\%$	3.6
50.0%	0.0000 ± 0.0000	$99.99\% \pm 0.01\%$	9.2	0.1018 ± 0.0088	$32.15\% \pm 5.87\%$	5.6

Table: Noise robustness benchmark comparison ($\nu_{\text{true}} = 0.1500$).

Backup: Shakura–Sunyaev Derivation

1. Expand inner derivative: $\partial_R(fR^{1/2}) = R^{1/2}\partial_R f + \frac{1}{2}R^{-1/2}f$
2. Multiply by $R^{0.5}$: $R^{1/2}[\partial_R(fR^{1/2})] = R\partial_R f + \frac{1}{2}f$
3. Apply outer derivative: $\partial_R(R\partial_R f + \frac{1}{2}f) = R\partial_R^2 f + \frac{3}{2}\partial_R f$
4. Multiply by $3/R$: $\frac{3}{R}(R\partial_R^2 f + \frac{3}{2}\partial_R f) = 3\partial_R^2 f + \frac{9}{2R}\partial_R f$

Backup: Geometrized Units ($G = c = 1$)

Definition

- $G = c = 1$.
- Length, time, and mass share the same unit M :

$$R_g = \frac{GM}{c^2} \rightarrow M, \quad t_g = \frac{GM}{c^3} \rightarrow M$$

- Scale-free formulation of General Relativity.

Multiply dimensionless quantities by characteristic scales:

- $R_{\text{phys}} = R \cdot \left(\frac{GM}{c^2}\right) \quad [\text{cm}]$
- $t_{\text{phys}} = t \cdot \left(\frac{GM}{c^3}\right) \quad [\text{s}]$
- $\nu_{\text{phys}} = \nu \cdot \left(\frac{GM}{c}\right) \quad [\text{cm}^2/\text{s}]$
- **Example ($M = 10 M_{\odot}$ BH):**
 - $1 M \approx 14.8 \text{ km}$
 - $1 M \approx 4.93 \times 10^{-5} \text{ s}$